

SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 2

SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING **COURSE CODE: ECA09**

Code of Conduct:

I _Kapil S P (192412182) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary actionn

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Experiment 14: Blackman-Tukey Spectral Estimation

Aim

To estimate the spectrum of a 300 Hz signal using the Blackman-Tukey method for different autocorrelation window lengths (32, 64, and 128), and to study spectral resolution.

Given Data

- Signal frequency, ($f_0 = 300$) Hz
- Sampling frequency, ($F_s = 3000$) Hz
- Autocorrelation window lengths, ($M = 32, 64, 128$)

Requirements

- MATLAB software
- Computer system

Theory

Blackman-Tukey spectral estimation is a non-parametric method used to estimate the Power Spectral Density (PSD) of a signal.

The steps are:

1. Generate or obtain the signal.
2. Compute its autocorrelation sequence.
3. Apply a window to the autocorrelation sequence.
4. Take the Fourier Transform of the windowed autocorrelation.
5. Plot the estimated spectrum.

MATLAB Program

% Q14: Blackman-Tukey Spectral Estimation

clc;

clear;

close all;

% Given data

f0 = 300; % Signal frequency in Hz

Fs = 3000; % Sampling frequency in Hz

N = 1024; % Number of signal samples

n = 0:N-1;

t = n/Fs;

% Test signal: 300 Hz sinusoid with small noise

*x = sin(2*pi*f0*t) + 0.15*randn(1,N);*

% Autocorrelation window lengths

M_values = [32 64 128];

NFFT = 4096;

figure;

for k = 1:length(M_values)

M = M_values(k);

% Autocorrelation sequence for lags -M to +M

lags = -M:M;

rxx = zeros(size(lags));

for i = 1:length(lags)

lag = lags(i);

```

    if lag >= 0
        rxx(i) = sum(x(1:N-lag) .* x(1+lag:N)) / N;
    else
        rxx(i) = sum(x(1-lag:N) .* x(1:N+lag)) / N;
    end
end

% Blackman window applied to autocorrelation sequence
L = length(rxx);
m = 0:L-1;
w = 0.42 - 0.5*cos(2*pi*m/(L-1)) ...
    + 0.08*cos(4*pi*m/(L-1));

rxx_windowed = rxx .* w;

% Blackman-Tukey spectral estimate
Pxx = abs(fftshift(fft(rxx_windowed, NFFT)));
Pxx = Pxx / max(Pxx);    % Normalize spectrum

% Frequency axis
f = (-NFFT/2:NFFT/2-1)*(Fs/NFFT);

% Plot only positive-frequency spectrum
subplot(3,1,k);
plot(f(f >= 0), Pxx(f >= 0), 'LineWidth', 1.5);
grid on;
xlim([0 800]);
xlabel('Frequency (Hz)');
ylabel('Normalized PSD');
title(['Blackman-Tukey Spectrum: Autocorrelation Window Length M = ', num2str(M)]);

end

```

Procedure

1. Set the signal frequency as 300 Hz and sampling frequency as 3000 Hz.
2. Generate the sinusoidal signal with small random noise.
3. Select autocorrelation window lengths as 32, 64, and 128.
4. Compute the autocorrelation sequence for each window length.
5. Apply the Blackman window to the autocorrelation sequence.
6. Perform FFT on the windowed autocorrelation sequence.
7. Plot the normalized power spectrum.
8. Compare the spectral peaks obtained for all three window lengths.

Observation Table

Window Length (M)	Approximate Resolution (Δf = F_s/M)	Observation
32	93.75 Hz	Broad peak around 300 Hz; low resolution
64	46.88 Hz	Sharper peak; improved resolution
128	23.44 Hz	Narrowest peak near 300 Hz; best resolution

Result

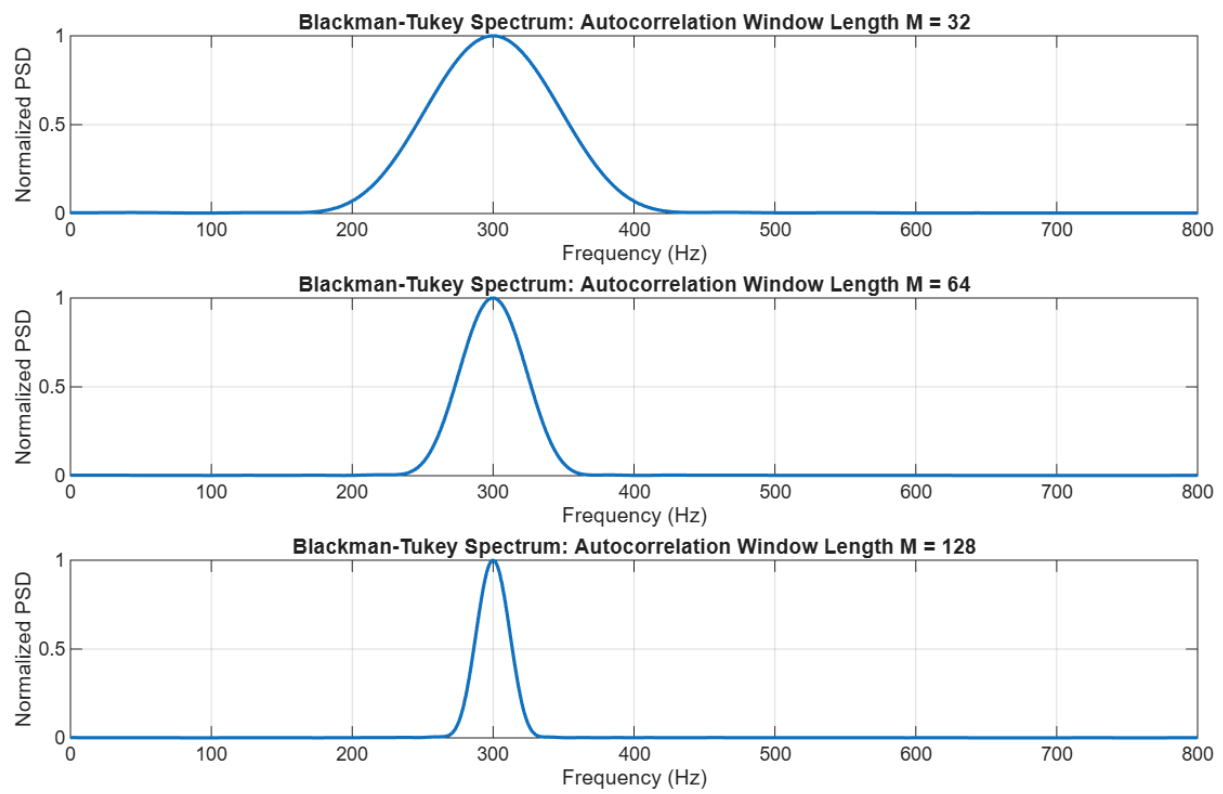
The Blackman-Tukey spectral estimation method was successfully simulated for autocorrelation window lengths of 32, 64, and 128.

The spectrum shows a major peak near the input signal frequency of **300 Hz**.

- For (M=32), the peak is broad and the resolution is low.
- For (M=64), the peak becomes sharper.
- For (M=128), the peak is narrowest and the frequency resolution is highest.

Therefore, increasing the autocorrelation window length improves spectral resolution.

OUTPUT



Advantages

1. It provides a smooth spectral estimate.
2. It reduces spectral leakage by using a window.
3. It is useful for noisy signals.
4. It allows control of the trade-off between smoothness and resolution.
5. It is simple to implement using autocorrelation and FFT.

Limitations

1. A small window length gives poor frequency resolution.
2. A large window length can increase spectrum variation.
3. The result depends on the selected window type.
4. It requires more computation for larger autocorrelation lengths.
5. The Blackman window reduces sidelobes but broadens the main spectral peak.

Conclusion

The Blackman-Tukey method estimates the power spectrum by applying a Blackman window to the autocorrelation sequence. Among the tested values, ($M=128$) gives the best spectral resolution because it produces the sharpest peak at 300 Hz.

```

2   clc;
3   clear;
4   close all;
5
6   % Given data
7   f0 = 300;           % Signal frequency in Hz
8   Fs = 3000;          % Sampling frequency in Hz
9   N = 1024;           % Number of signal samples
10  n = 0:N-1;
11  t = n/Fs;
12
13  % Test signal: 300 Hz sinusoid with small noise
14  x = sin(2*pi*f0*t) + 0.15*randn(1,N);
15
16  % Autocorrelation window lengths
17  M_values = [32 64 128];
18  NFFT = 4096;
19
20  figure;|
21
22  for k = 1:length(M_values)
23
24      M = M_values(k);
25
26      % Autocorrelation sequence for lags -M to +M
27      lags = -M:M;
28      rxx = zeros(size(lags));
29
30      for i = 1:length(lags)
31          lag = lags(i);
32
33          if lag >= 0
34              rxx(i) = sum(x(1:N-lag) .* x(1+lag:N)) / N;
35          else
36              rxx(i) = sum(x(1-lag:N) .* x(1:N+lag)) / N;
37          end
38      end
39
40      % Blackman window applied to autocorrelation sequence
41      L = length(rxx);
42      m = 0:L-1;
43      w = 0.42 - 0.5*cos(2*pi*m/(L-1)) ...
44          + 0.08*cos(4*pi*m/(L-1));
45
46      rxx_windowed = rxx .* w;
47
48      % Blackman-Tukey spectral estimate
49      Pxx = abs(fftshift(fft(rxx_windowed, NFFT)));
50      Pxx = Pxx / max(Pxx); % Normalize spectrum
51
52      % Frequency axis
53      f = (-NFFT/2:NFFT/2-1)*(Fs/NFFT);
54
55      % Plot only positive-frequency spectrum
56      subplot(3,1,k);
57      plot(f(f >= 0), Pxx(f >= 0), 'Linewidth', 1.5);
58      grid on;
59      xlim([0 800]);
60      xlabel('Frequency (Hz)');
61      ylabel('Normalized PSD');
62      title(['Blackman-Tukey Spectrum: Autocorrelation Window Length M = ', num2str(M)]);
63  end

```